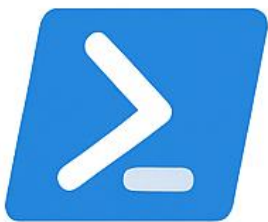




PHASE 2 – INSTALLING THE DOMAIN CONTROLLER

Term	Definition
Domain Controller	Server that stores the Active Directory database and responds to authentication and authorization in domain.
OU (Organizational Unit)	Logical container within AD for grouping objects, containers, and policies, via an object.
UPN Suffix	Part of the User Principal Name (e.g. @yourdomain.com) which identifies domain for login
DSRM	Directory Services Restore Mode, as a password used to boot into recovery mode.
Promotion	Process of installing the AD DS role and converting a server into a Domain Controller



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DSRM	Directory Services Restore Mode; password used to start in recovery mode.
Promotion	Process of installing the AD DS role and converting a server into a Domain Controller.

PHASE 2 – INSTALLING THE DOMAIN CONTROLLER

Prerequisites

- Local VM running on VirtualBox with **Windows Server 2022**
- Network connectivity between local environment and Azure (configured in **Phase 3**)

1. Download Windows Server 2022

Download ISO file:

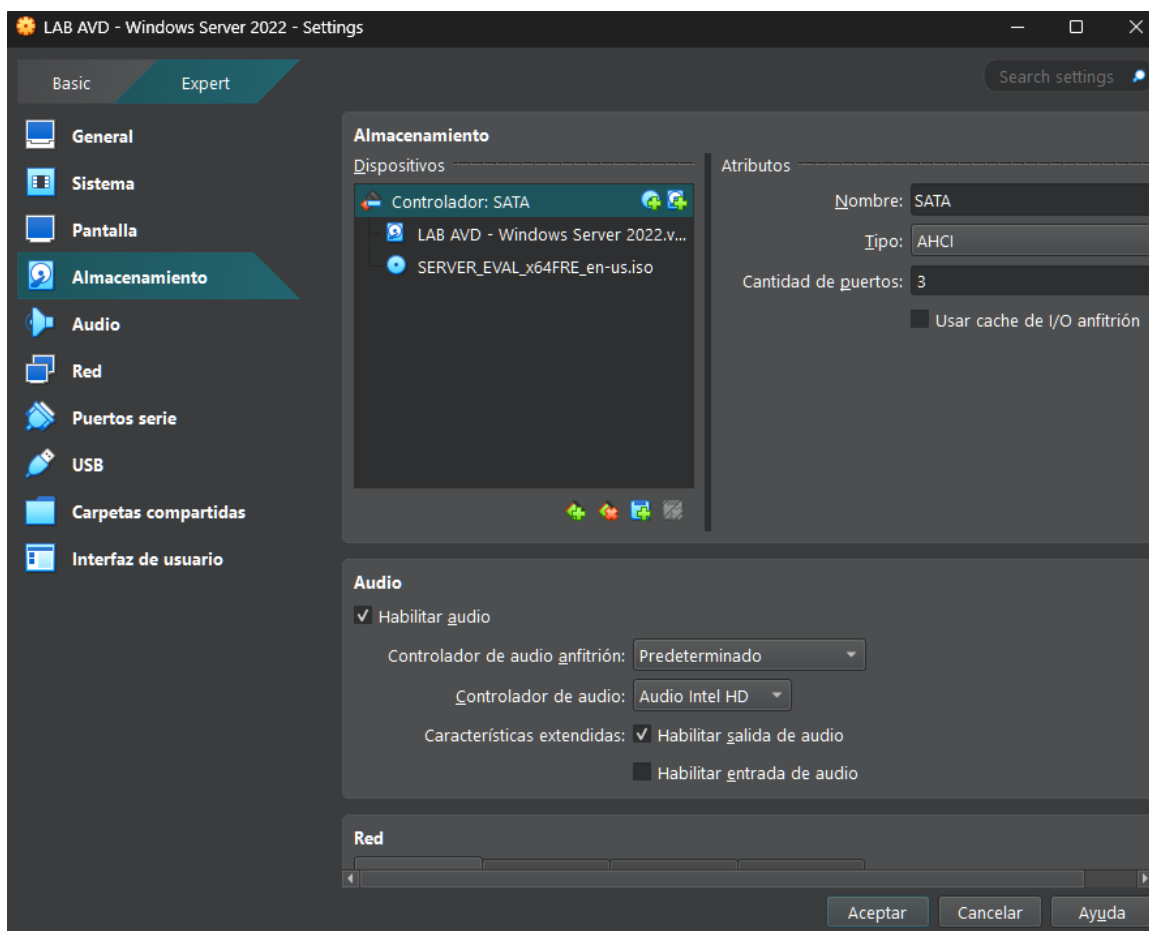
SERVER_EVAL_x64FRE_en-us.iso

2. VirtualBox VM Configuration

Initial settings:

- **Name:** WS2022-DC01
- **RAM:** 4096 MB
- **CPU:** 2 cores
- **Disk:** 60 GB
- **Video RAM:** 128 MB
- **Mount ISO:** SERVER_EVAL_x64FRE_en-us.iso as CD/DVD drive 1

 **Important:** Make sure to **uncheck** "Unattended installation"



Verify video settings:

- Go to VM settings
- Section: *Display*
- Set **Video RAM**: 128 MB

3. Windows Server Installation

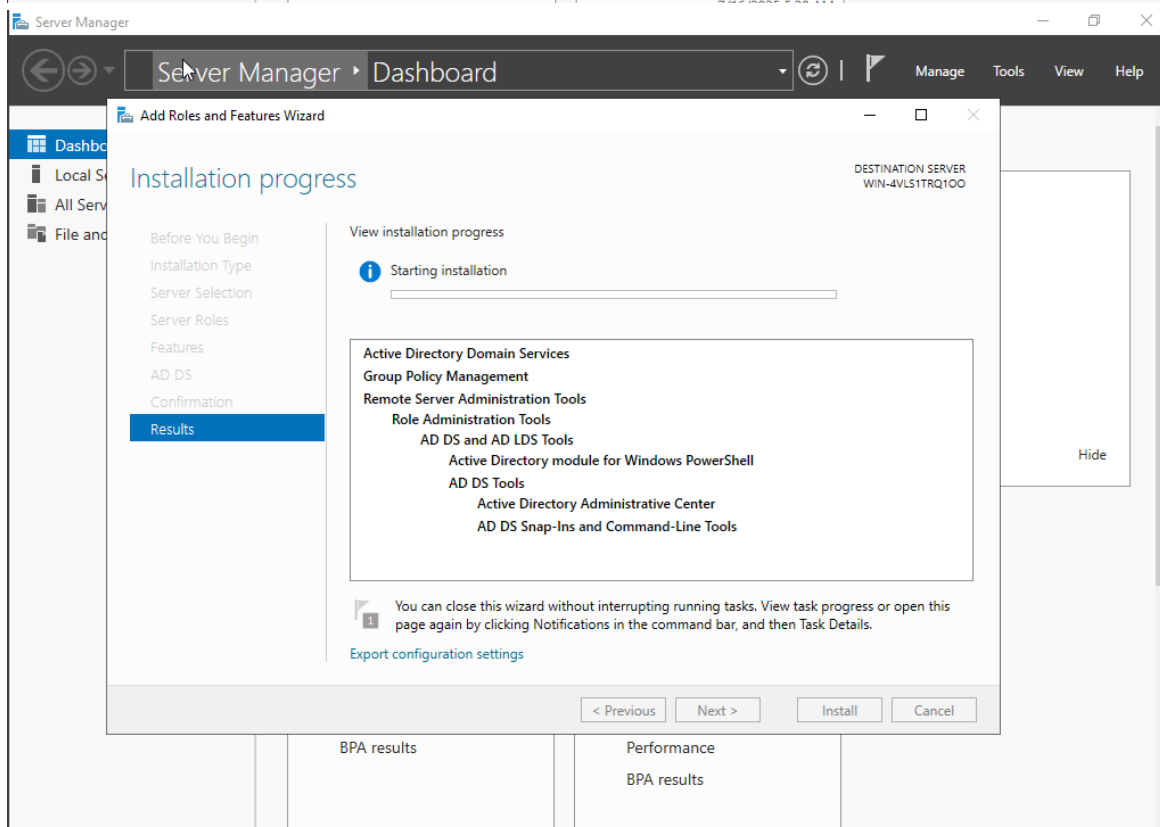
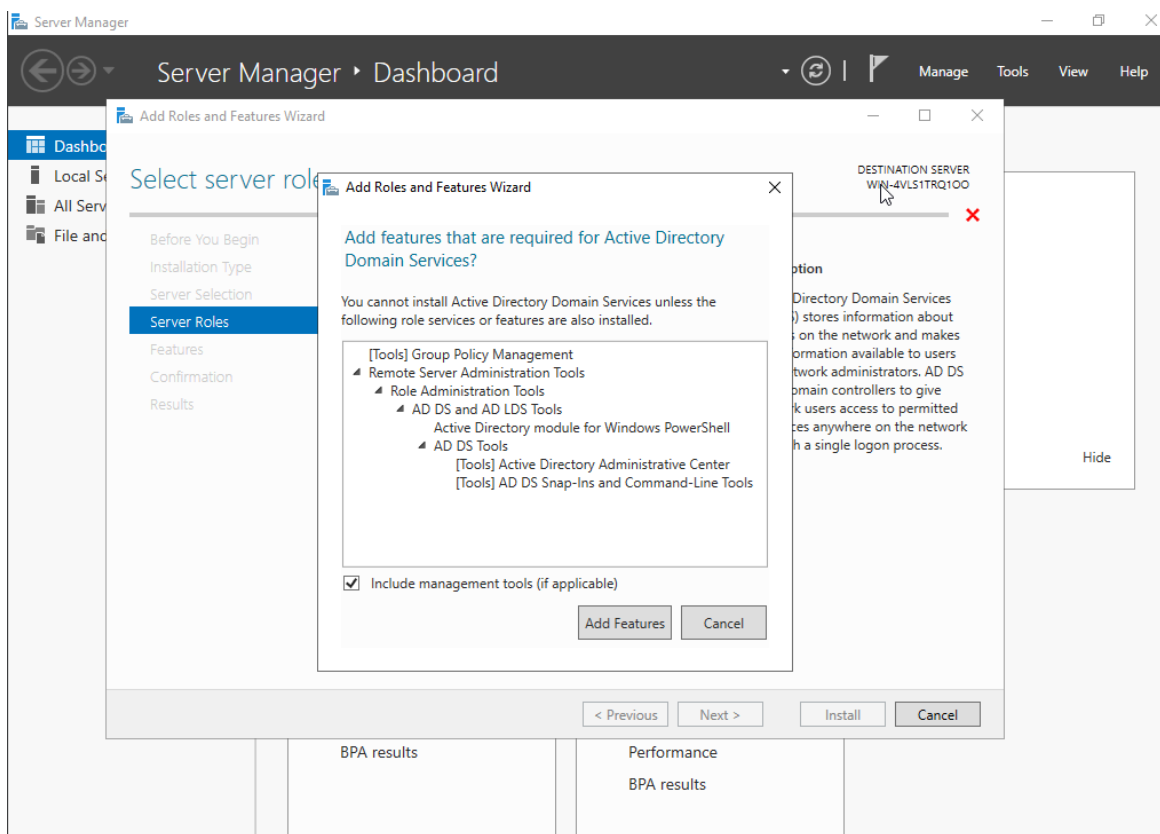
Steps:

- Start the VM
- Select edition: choose one with **Desktop Experience**
- Accept license terms
- Installation type: **Custom install**
- Disk: select available disk (60 GB)
- Complete installation

4. Post-installation Configuration

4.1 Install Active Directory Domain Services

- Open **Server Manager**
- Go to **Add Roles and Features**
- Select **Active Directory Domain Services**
- Complete role installation



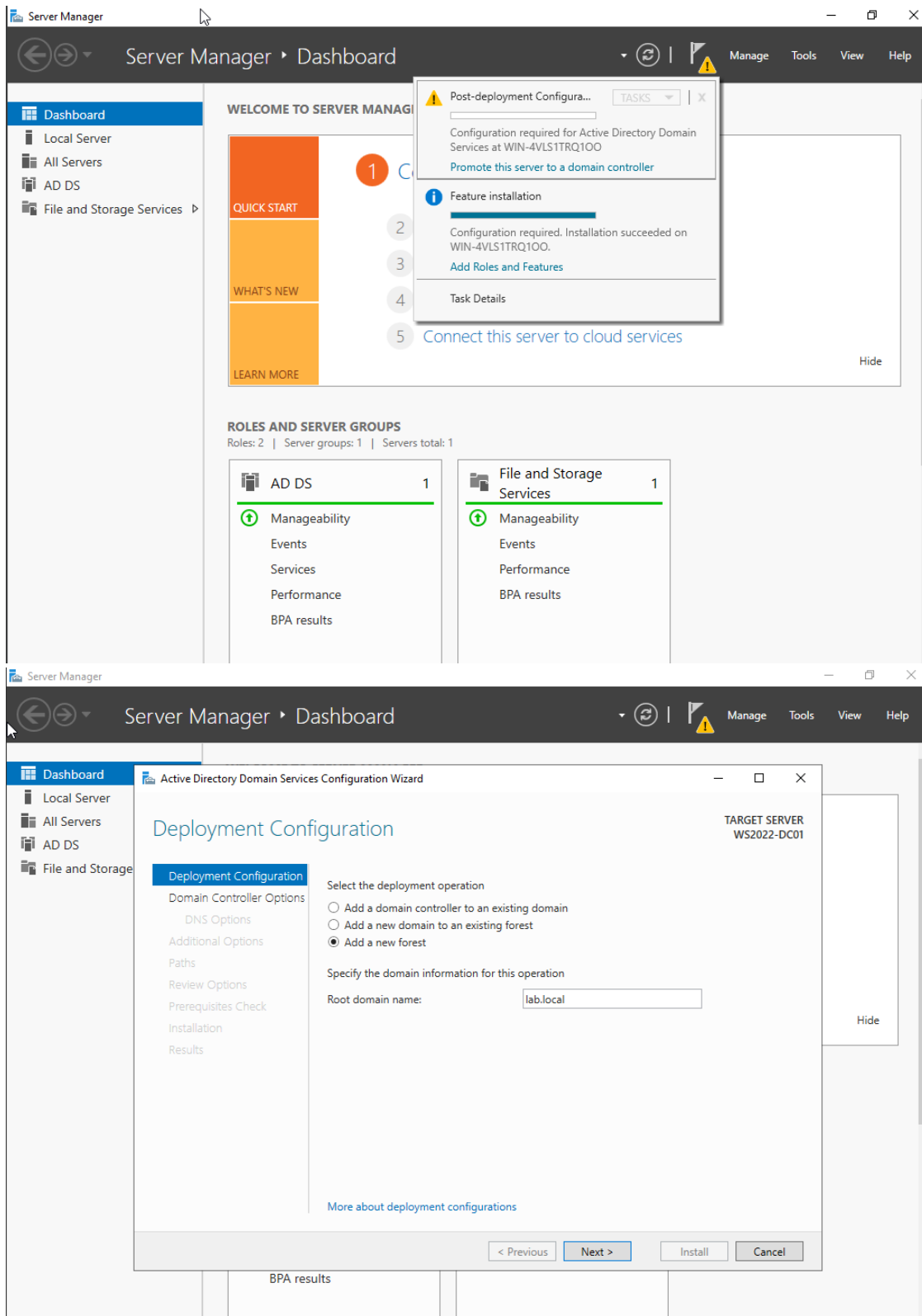
4.2 Rename the Server

- In Server Manager → *Local Server*
- Click on the computer name
- Change to: WS2022-DC01
- Restart the server

4.3 Promote to Domain Controller

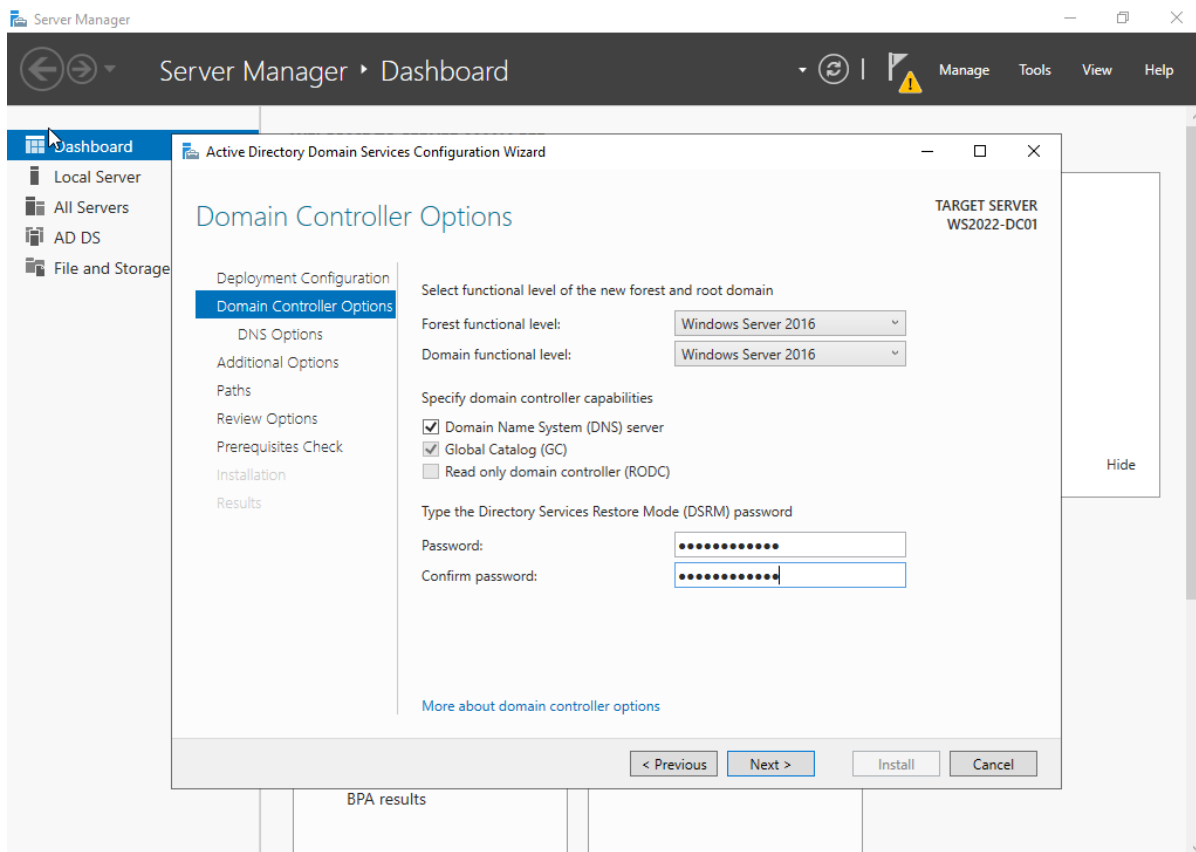
- In Server Manager, click the **flag icon** (notifications)
- Select: **Promote this server to a domain controller**
- **Domain Configuration:**
 - Select: **Add a new forest**
 - Root domain name: `lab.local`

- Forest and Domain Functional Level: **default**

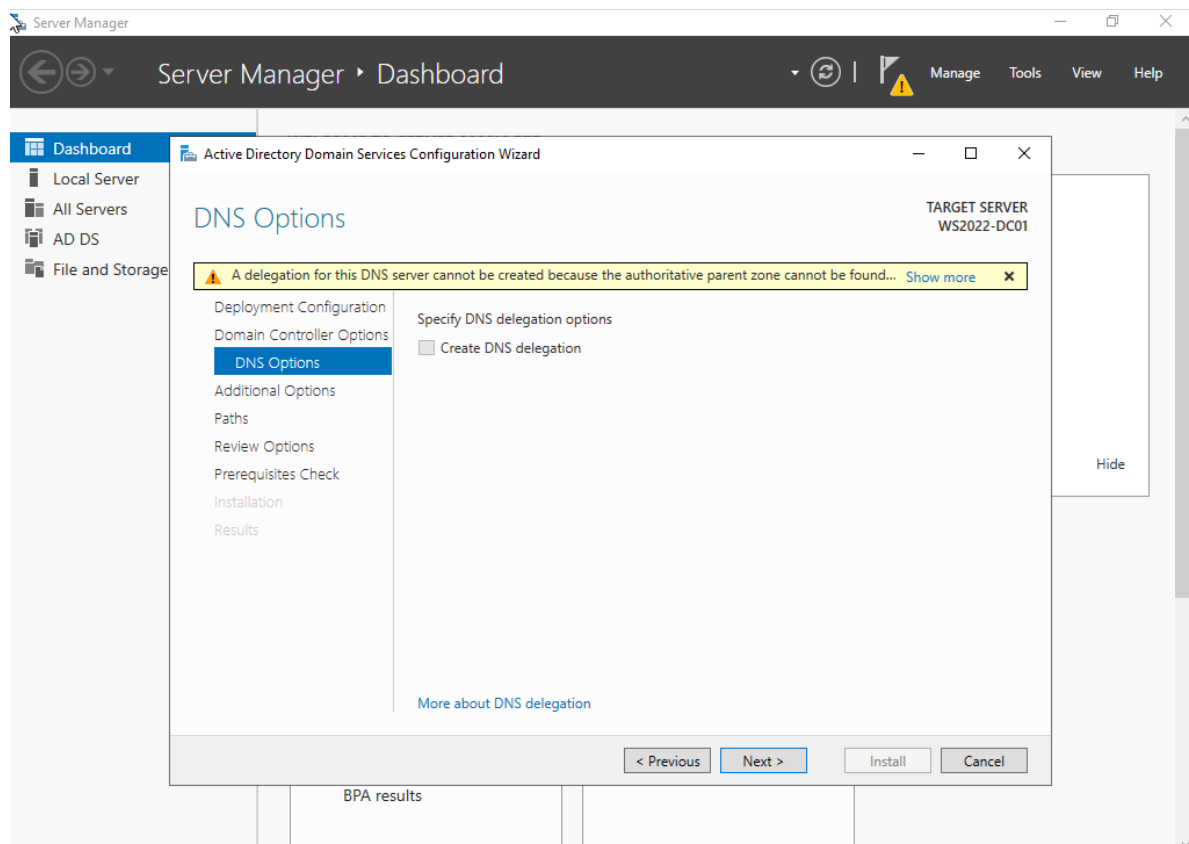


Domain Controller Options:

- Keep defaults
- Set DSRM password: <your password>



c. DNS Options: Continue with default settings



d. Additional Options: NetBIOS domain name: LAB

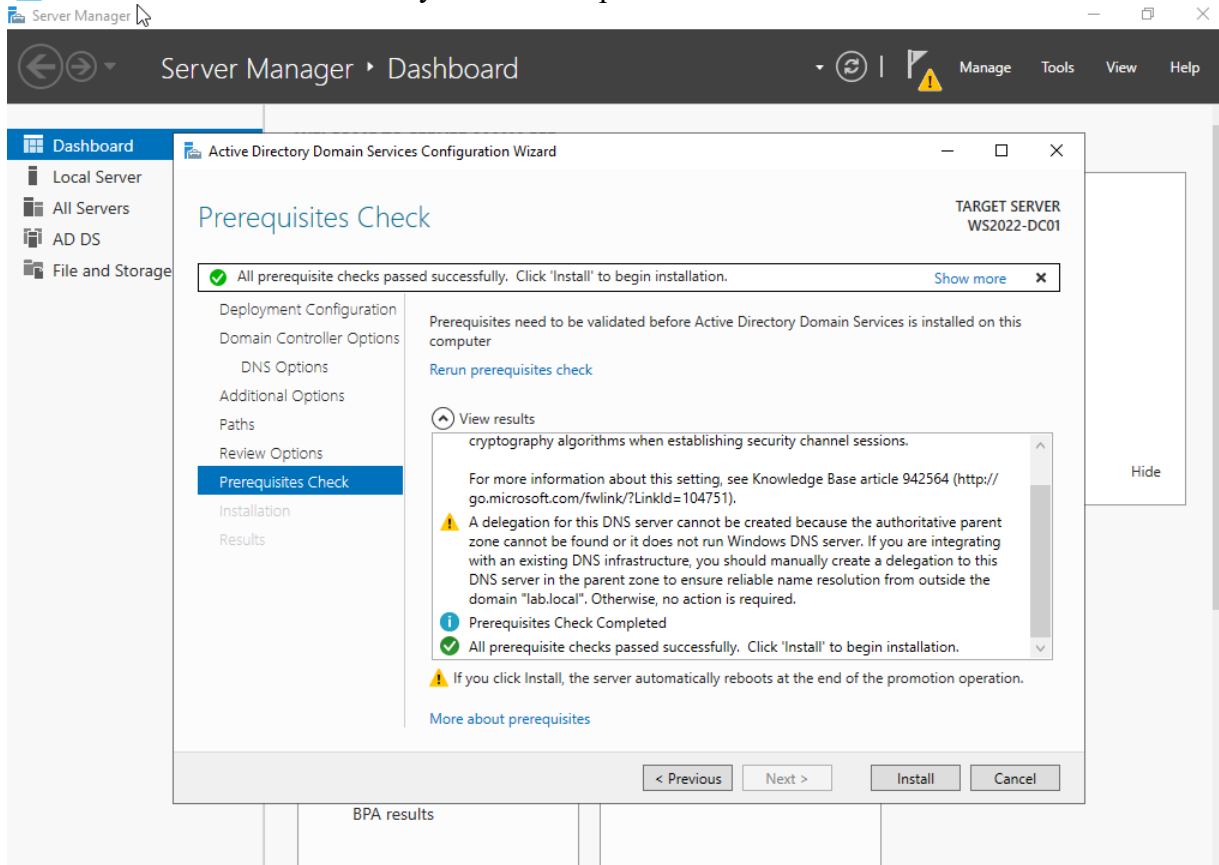
e. **Paths:** keep defaults

Review Options: confirm all settings

Prerequisites Check: wait for validation

Install: proceed with installation

The server will automatically reboot after promotion.



5. Verify Installation

After reboot:

1. log in with domain admin account:
 - **Username:** LAB\Administrator
 - **Password:** <your password>
2. **Verify AD Services:**
 - Open **Active Directory Users and Computers**
 - Check that domain lab.local is listed
 - Confirm default OUs are present
3. **Verify DNS:**
 - Open **DNS Manager**
 - Confirm DNS zones for lab.local exist

6. Prepare for Azure Sync

6.1 Configure Custom UPN Suffix

- Open **Active Directory Domains and Trusts**
- Right-click the root node → *Properties*
- In **UPN Suffixes** tab:
 - Add: lucasvlabs.site
 - Click **Add**, then **Apply** and **OK**

6.2 Create OU for Sync

- Open **Active Directory Users and Computers**
- Right-click on lab.local → New → **Organizational Unit**
- Name: SyncLab
- Click OK

6.3 Create Test User

- In **Active Directory Users and Computers**, go to SyncLab
- Right-click → New → **User**

User Details:

- First name: Lucas
- Last name: Vercellini
- User logon name: lucasv
- UPN suffix: select @lucasvlabs.site
- Set password

User Options:

- ☒ Password never expires
- ☒ User cannot change password
- Finish

El Domain Controller está listo para la sincronización con Azure Entra ID (Fase 3).